

## Education

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- 2021–2027 **Ph.D. in Mathematics**, *University of Oklahoma*  
Advisor: Peter Patzt  
GPA: 3.82
- 2019–2021 **M.S. in Mathematics**, *American University of Beirut*  
Advisor: Hazar Abu-Khuzam  
Thesis Title: Structure of certain rings with conditions on nonperiodic elements and certain Von-Neumann  $\Pi$ -regular rings with prime centers  
GPA: 3.95
- 2015–2018 **B.S. in Mathematics**, *Lebanese University*  
GPA: 3.56

## Research Papers

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1. **Double Steinberg coinvariants for special linear groups**, with David Chan, Alexander Kupers, Robin J. Sroka, and Matthew Scalamandre, arxiv:2609.39991
2. **Codimensions one and two cohomology of Hecke congruence subgroups** (submitted), with Jeremy Miller, arXiv:2606.23519
3. **Top-dimensional rational cohomology of the congruence subgroup  $\Gamma_{0,n}^+(p)$**  (submitted), arXiv:2605.23526

## Honors, Awards, and Fellowships

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### University of Oklahoma:

- 2026
- **MGSA Good Mentor Award**
  - **Thomas J. Hill Graduate Fellowship in Mathematics** (\$2,000)
  - **T. Jay Thompson Scholarship Award** (\$1,750)
  - **Armstrong Memorial Scholarship Award** (\$3,000)
  - **SHS-OG&E Energy Corp Scholarship Award** (\$1,000)
  - **Dodge Family Completion Fellowship Award** (\$12,200)
  - **Dodge Family College of Arts and Sciences Travel Assistance Program** (\$750)
- 2025
- **David and Judi Proctor Scholarship** (\$2,750)
  - **Robert Lee Loshbough Endowed Scholarship Fund** (\$2,750)
  - **Arts and Sciences - 31171** (\$1,000)
  - **Dodge Family College of Arts and Sciences Travel Assistance Program** (\$750)

- 2024 ○ **Summer Graduate Fellowship** (\$4,000)
- **Dodge Family College of Arts and Sciences Travel Assistance Program** (\$600)
- 2023 ○ **Conrad and Ellen Masterson Fellowship in Mathematics** (\$2,000)
- **Dodge Family College of Arts and Sciences Travel Assistance Program** (\$650)
- 2022 **C. Eugene Springer Mathematics Scholarship** (\$2,000)
- 2021-2025 **Fellowship from the Mathematics department** (\$16,000)
- 2021-Present **Graduate Teaching Assistantship**

American University of Beirut:

2029-2021 **Graduate Assistantship**

## Research Talks

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- Aug. 2026 **Cohomology of Hecke congruence subgroups (lightning talk)**, *Summer 2026 Wasatch Topology Conference, Utah*
- Jun. 2026 **On the rational (co)homology of Hecke congruence subgroup**, *Topology Students Workshop, Vanderbilt University*
- May 2026 **Cohomology of congruence subgroup  $\Gamma_{0,n}(p)$** , *International Seminar on Quantization, Geometry, and Dynamics, University of Texas Rio Grande Valley (virtual)*
- Apr. 2026 **Top-dimensional cohomology of congruence subgroup  $\Gamma_{0,n}(p)$** , *Topology seminar, Purdue University*
- Feb. 2026 **Top-dimensional cohomology of congruence subgroup  $\Gamma_{0,n}(p)$** , *Topology seminar, University of Michigan*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup  $\Gamma_{0,n}(p)$** , *Geometry and Topology Seminar, University of Oklahoma*
- Jul. 2025 **Double Steinberg Coinvariants**, *Scissors Congruence and K-theory Conference, University of Pennsylvania*

## Poster Presentations

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- Jan. 2027 **Cohomology of Hecke congruence subgroups**, *AWM Workshop at the Joint Mathematics Meetings (JMM) 2027*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup  $\Gamma_{0,n}(p)$** , *Perspectives on the cohomology of general linear groups, University of Münster*

## Expository Talks

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- Nov. 2025 **Borel–Serre duality**, *Student Topology Seminar, University of Oklahoma*
- Nov. 2023 **High Dimensional Cohomology of Arithmetic Groups**, *The 2023 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference, Binghamton University*

Dec. 2022 **Introduction to Spectral Sequences** , *Student Presentation Seminar, University of Oklahoma*

## Teaching Experience

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### Instructor of Record:

- SP26 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA25 **Functions and Modeling for Business, Life and Social Sciences.** (large section  $\sim 80$  students), *University of Oklahoma*
- SP24 **Calculus I for Business, Life and Social Sciences**, *University of Oklahoma*
- FA23 **Functions and Modeling for Business, Life and Social Sciences.**, *University of Oklahoma*
- SU23 **Calculus & Analytics Geometry IV**, *University of Oklahoma*
- SP23 **Functions and Modeling for Business, Life and Social Sciences.**, *University of Oklahoma*
- SU22 **Calculus & Analytics Geometry I**, *University of Oklahoma*

### Discussion Leader:

- SP25 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA24 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA22 **Differential & Integral Calculus I**, *University of Oklahoma*
- SP22 **Calculus and Analytics Geometry I**, *University of Oklahoma*

### Grader:

- SP25 **Topology II**, *University of Oklahoma*
- FA24 **Topology I**, *University of Oklahoma*
- FA22 **Introduction to Analysis I**, *University of Oklahoma*
- FA21 **Calculus & Analytics Geometry IV**, *University of Oklahoma*
- FA21 **Differential & Integral Calculus III (HONORS)**, *University of Oklahoma*
- FA20 **Rings & Modules I**, *American University of Beirut*
- FA19, SP20 **Introduction to Analysis I**, *American University of Beirut*

## Service and Outreach

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### University of Oklahoma:

- 2025-2026 **Treasurer of the University of Oklahoma AWM Chapter**
- Spring 2025 **Mentor, Directed Reading Program**, On the Proof of Abel and Ruffini
- 2024–2025
  - **Graduate Student Senate**
  - **Outreach of the University of Oklahoma AWM Chapter**
  - **Co-organizer of the University of Oklahoma Graduate Math Fest** (an annual Graduate Student Conference at the University of Oklahoma)

- 2023-2024   ○ Co-founder & President of the University of Oklahoma AWM Chapter  
               ○ Co-founder of the AWM Mentorship Program for undergraduate students  
               ○ Secretary of the Math Graduate Student Association
- 2022-2023   Member of the Graduate Advocacy Committee

## Conferences and Workshops attended

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- May. 2026   **Configuration spaces summer school**, *University of Michigan*
- Jun. 2025   **Polylogarithms, homology of linear groups, and Steinberg modules**, *SwissMAP Research*, Station in Les Diablerets
- May. 2025   **Summer school: Homological aspects of groups and their classifying spaces**, *SwissMAP Research*, Station in Les Diablerets
- Feb. 2025   **Redbud Topology Conference Spring 2025**, *University of Arkansas*
- Jul. 2024   **Collaborative Workshop in K-theory and Scissors Congruence**, *Vanderbilt University*
- Oct. 2023   **The South Central Topology Conference II**, *New Mexico State University*
- Oct. 2023   **Redbud Topology Conference Fall 2023**, *University of Arkansas*
- Jul. 2023   **Stability in Topology, Arithmetic, and Representation Theory 2023**, *Purdue University*
- Jun. 2023   **Master class : Homology of groups and functors**, *Université de Lille*
- April 2023   **Redbud Topology Conference Spring 2023**, *University of Oklahoma*
- Feb. 2023   **The South Central Topology Conference II**, *Texas State University*
- Nov. 2022   **The 2022 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference**, *Binghamton University*
- Oct. 2022   **Redbud Topology Conference Fall 2022**, *University of Arkansas*
- April 2022   **Texas-Oklahoma Representations and Automorphic forms XI**, *Oklahoma State University*